14. RAW SOCKETS to implement packet sniffing.

Aim: To write a code using raw sockets for packet sniffing

Code:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP
def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst

        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif
            protocol == 17:
            protocol_name = UDP"
    else:
        protocol_name = "Unknown Protocol"

    print(f" Protocol : {protocol_name}")
    print(f" Source IP : {src_ip}")
    print( f" Destination IP: {dst_ip} ")
    print ("_" * 50)

sniff (iface = "wifi", prn = packet_callback, filter = "ip",
       store = 0)
```

Output

C:\Users\Vishnu\Downloads>python packet.py

Protocol : TCP

Source IP : 57.15.141.117

Destination IP : 192.168.1.6

Protocol : TCP

source IP : 192.168.1.6

Destination IP : 57.155.141.117

Protocol : UDP

Source IP : 192.168.1.6

Destination IP : 192.168.1.1

Protocol : UDP

source IP : 198.168.1.1

Destination IP : 192.168.1.6

Result:

Thus the raw sockets is used to implement packet sniffing successfully.